📋 Internship Project — Week 3 & 4

Executive Business Report: End-to-End Sales Forecasting & Demand Intelligence System

Executive Summary This report outlines the development and findings of our new AI-driven Sales Forecasting and Demand Intelligence system. By leveraging 4 years of historical transaction data—and successfully integrating an external supplementary dataset to perform multi-source analysis—we built a system that predicts future product demand with high accuracy, automatically flags anomalous sales spikes or drops, and segments our product catalog based on distinct purchasing behaviors. This enables the supply chain and finance teams to optimize inventory levels, minimize overstock costs, and ensure high-demand items are consistently available, driving both operational efficiency and revenue growth.

Key Findings from EDA and Forecasting

- **Revenue Drivers:** The "Technology" category remains our highest revenue generator, consistently outperforming Furniture and Office Supplies.
- **Regional Consistency:** The West region has demonstrated the most consistent and stable sales growth year-over-year.
- **Seasonality:** There is a massive, predictable sales spike occurring every November and December, aligning with holiday shopping.
- **Forecasting Model Selection:** After evaluating SARIMA, Facebook Prophet, and XGBoost, **Prophet** was selected as the production model due to its robust handling of our strong yearly seasonality, providing the lowest Mean Absolute Percentage Error (MAPE).

3-Month Sales Forecast: Based on our Prophet forecasting model, the next 3 months indicate steady baseline growth with expected seasonal fluctuations. The model provides an 80% confidence interval. We observe the strongest upcoming growth trend in the Technology segment within the West region.

Top 3 Anomalies Detected (Multi-Source Approach): By successfully merging our internal sales data with an external supplementary dataset (vgsales.csv), we performed advanced multi-variate anomaly detection. Using Z-Score and Isolation Forest techniques on these combined datasets, we flagged several significant historical anomalies:

1. **Late November Spikes (Recurring):** Sales volume deviated by over +3 standard deviations. *Likely Cause:* Black Friday and Cyber Monday promotional events.

2. **Mid-January Drop (Year 2):** A sharp decline in sales below expected post-holiday levels. *Likely Cause:* Severe supply chain bottleneck or extreme weather disrupting deliveries and suppressing immediate re-orders.
3. **Unexpected Mid-Year Spike (Year 3):** An isolated spike in the 'Office Supplies' category during July. *Likely Cause:* A major bulk B2B contract signed or a specific back-to-school regional campaign.

Product Demand Segmentation Using K-Means clustering on product sub-categories, we identified four distinct demand segments:

- **Segment 1: High Volume, Stable Demand (e.g., Paper, Binders).**
  - *Stocking Strategy:* Maintain high safety stock levels and utilize automated, regular reordering to prevent stockouts.
- **Segment 2: High Volatility, High Value (e.g., Copiers, Machines).**
  - *Stocking Strategy:* Implement Just-In-Time (JIT) inventory management or vendor drop-shipping to minimize expensive warehousing costs.
- **Segment 3: Low Volume, Declining (e.g., specific older tech accessories).**
  - *Stocking Strategy:* Halt reorders, phase out inventory, and apply aggressive discounts to clear warehouse space.
- **Segment 4: Growing Demand (Star Products).**
  - *Stocking Strategy:* Prioritize warehouse allocation and increase buffer stock to capture all upside potential without risking stockouts.

Business Recommendations

1. **Automate Holiday Procurement:** Given the massive, predictable spikes in Q4 identified by our models, Supply Chain should lock in vendor contracts and secure warehouse space by late Q3 to avoid premium last-minute logistics costs.
2. **Implement Segment-Specific Inventory Policies:** Shift away from a one-size-fits-all inventory model. Immediately apply Just-In-Time ordering for the "High Volatility, High Value" cluster (Cluster 2) to free up working capital.
3. **Targeted Marketing for the West Region:** Since the West region shows the most consistent growth and strong forecast, allocate additional marketing budget to this region to capitalize on the established momentum and maximize ROI.

Risks and Limitations

- **External Shock Vulnerability:** The forecasting models rely heavily on historical patterns. They cannot accurately predict unprecedented macroeconomic shocks (e.g., a global pandemic, sudden massive inflation, or unforeseen supply chain blockades).